

At present, it is very difficult to discern research risk for institutions, as the available numbers of publication rate, journal impact factors, and citation counts, are individually highly exploitable and loosely correlative to their intended quantity of measure, but also have myriad interpretations that are all valid, leading to a high noise to signal ratio for determining academic risk and poor consensus amongst academics of what the story behind each number means. By contrast, the way the Liberata marketplace and metrics are constructed, if risk premium (perceived risk) is actually inaccurate to true risk, there is additional expected academic capital to be arbitrated from doing or avoiding peer review or replication for that undervalued or overvalued work respectively, allowing the system to self correct inaccurate pricing through the classic market forces that lead to price discovery.

7.3 Relative Performance and Risk Adjusted Performance

A simple relative performance measure for any manuscript is to take its expected returns for academic capital μ_m defined in section 6.5 as a ratio of the expected returns $\mu_{d(m)}$ of the average manuscript in m 's academic field d , as denoted by m 's primary domain, department, discipline and direction tags.

$$\text{Relative performance}_m = \rho_m = \frac{\mu_m}{\mu_{d(m)}} \quad (58)$$

This measure quantifies the gains in academic capital per unit time for the manuscript as a ratio to the field manuscripts' average, which is a useful and intuitive way to indicate the quality of a manuscript. By extension, the relative performance of a portfolio Π is the weighted sum of the relative performances.

$$\text{Relative performance}_\Pi = \rho_\Pi = \frac{1}{\sum_{m \in \Pi} s_m} \sum_{m \in \Pi} s_m \frac{\mu_m}{\mu_{d(m)}} \quad (59)$$

Where s_m denotes % shares the portfolio holds in manuscript m . The portfolio relative performance is a useful, intuitive way to represent the performance of any entity that can hold a portfolio of shares on manuscripts, such as individuals, institutions, etc.

Additionally, the performance of any portfolio Π can be measured in a risk adjusted way by using a redefined capital asset pricing model (CAPM) Sharpe [1964] where the risk free return is set to 0, equivalent to the assumption that there is no generation of academic capital outside of manuscripts, which is true by construction of the Liberata system. First, we define the risk adjusted excess return α_m for any manuscript. This quantity represents the extra return above expected for the riskiness of the manuscript.

$$\alpha_m = \mu_m - \beta_m \mu_{d(m)} \quad (60)$$

where β_m is the sensitivity of μ_m and $\mu_{d(m)}$, defined as below.

$$\beta_s = \frac{\text{Covariance}(\mu_m, \mu_{d(m)})}{\text{Variance}(\mu_{d(m)})} \quad (61)$$

By extension, the excess risk adjusted returns for a portfolio is the weighted sum of the excess risk adjusted returns for each manuscript within that portfolio.

$$\alpha_\Pi = \sum_{m \in \Pi} s_m \alpha_m = \sum_{m \in \Pi} s_m (\mu_m - \beta_m \mu_{d(m)}) \quad (62)$$

Risk adjusted relative performance then is taken to be for a single manuscript:

$$\text{Risk Adjusted Relative Performance}_m = \tilde{\rho}_m = \frac{\alpha_m}{\mu_{d(m)}} \quad (63)$$

and by extension, for a portfolio:

$$\text{Risk Adjusted Relative Performance}_\Pi = \tilde{\rho}_\Pi = \sum_{m \in \Pi} \frac{\alpha_m}{\mu_{d(m)}} \quad (64)$$

These metrics are a simple but powerful way for Liberata to quantify the relative performance of any manuscript or portfolio of manuscripts that is field agnostic.

8 Distribution Metrics

The distributions of shares on manuscripts, academic domains, departments, disciplines, and directions, as well as the authors, peer reviewers, replicators, institutions, geographic regions, time periods that hold those shares, make for a rich tapestry of distribution metrics in addition to the ones already mentioned in prior sections.

8.1 Author Contribution Distributions

It is expected that different academic fields will have different natural distributions for authors. As more manuscripts accumulate on Liberata, archetype histograms and distributions can be formed for author shares on different academic fields. In addition, because valid distributions can be formed on any set of manuscripts, distributions can be made for any valid portfolio of manuscripts as defined in section 6.

With the author distributions, we can calculate the inequality of author share splits using metrics with similar formulations to section 6.8. The typical author share distribution by HHI for a given field d is:

$$\text{HHI author shares}(d) = \frac{1}{|M_d|} \sum_{m \in M_d} \sum_{a \in A_m} s_{a,m}^2 \quad (65)$$

This quantity can be compared to the HHI for a portfolio or a single manuscript to check for anomalies between an author's portfolio (eq. (66)) or a share distribution of a single manuscript (eq. (67)) compared to that author's academic field.

$$\text{HHI author shares}(\Pi) = \frac{1}{|M_\Pi|} \sum_{m \in M_\Pi} \sum_{a \in A_m} s_{a,m}^2 \quad (66)$$

$$\text{HHI author shares}(m) = \sum_{a \in A_m} s_a^2 \quad (67)$$

Then, the HHI difference (HHIRD) measures discrepancy between the HHI of the field and that of the manuscript or portfolio in a single number.

$$\text{HHID} = |\text{HHI}_\Pi - \text{HHI}_d| \quad (68)$$

Similarly to HHI, other distribution comparisons are possible by taking the Gini coefficient in eq. (42) or the entropy measure in eq. (43) with allocation weights by shares (eq. (44)) for the author shares in a field d and compare it to that of an arbitrary portfolio or manuscript by difference or ratio respectively. These metrics can call be used as an indicator of anomalous author share splits, which can indicate situations including but not limited to a budding scientific field unlike its parent field in work requirements, or a supervisor that is able to leverage power dynamics to gain shares, or paid for credit manuscript farm products, etc.

8.2 Population pyramids

One way of representing the demographics of academic fields or of academia in its entirety is to make population pyramids or histograms more generally of contributors based on $\mathbb{X}$. For these histograms, we choose $\log(\mathbb{X})$ as the measure axis and contributor count on the frequency axis. This choice reflects that researcher academic capitals like citation counts are likely to be Pareto distributed. Price [1976], Redner [1998] These histograms then, with appropriate number of bins, serve as a visualization of whether a field is growing, shrinking, or stagnant (refer fig. 3 for an illustrative of what the histograms might look like). If a field is growing in population, one would expect relatively many young researchers with low $\mathbb{X}$, whereas with a shrinking field, one expects relatively many older researchers with high $\mathbb{X}$, analogous to the interpretations for population pyramids in traditional demographics.

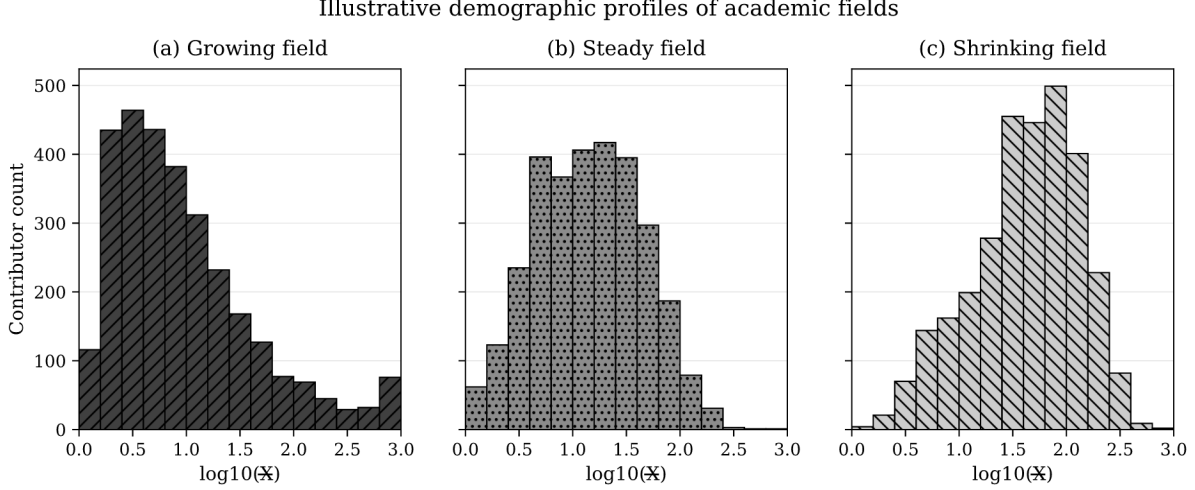


Figure 3: Illustrative demographic profiles of academic fields.

9 System Health Metrics

9.1 Academic Capital Growth Rates

As more academic works are produced, the total academic capital in the entire system is expected to grow. The rate of this growth can signal overall system health by measuring academic impact produced at a global level. This signal becomes more faithful to the actual health of the system as better quality control is implemented. This can be computed using eq. (34), where the portfolio becomes all the manuscripts that were published a unit time ago, $M' = M_{t-1}$, and $\Delta t = 1$. By default, Δt would be a year, following the convention in section 6.3.

9.2 Fair Market Price Shrinkage Rate

The fair market price of peer reviews and replications signal the actual costs associated with reviewing and replicating academic results. When it becomes easier to derive academic capital from quality control, the prices are expected to see a downward trend until plateauing. This metric would measure the rate of negative change in fair market price from eq. (54), eq. (55) for the whole system year-over-year. Let the change in fair market price of replications over the globe in a unit time period be denoted by $\Delta \bar{s}_r(t) = \bar{s}_r(t) - \bar{s}_r(t-1)$, where $\bar{s}_r(t) = \sum_{d \in D} \bar{s}_{r,d}^{(t)}$, and similarly, the change in fair market price of reviews be denoted as, $\Delta \bar{s}_p(t) = \bar{s}_p(t) - \bar{s}_p(t-1)$. Then, the global shrinkage rates are given by:

$$\bar{s}'_p = \frac{-\Delta \bar{s}_p(t)}{\Delta t} \quad (69)$$

$$\bar{s}'_r = \frac{-\Delta \bar{s}_r(t)}{\Delta t} \quad (70)$$

A weighted global shrinkage rate can be further computed by taking a weighted sum over both $\bar{s}'_p$ and $\bar{s}'_r$, weighted by $0 < \psi < 1$.

$$\bar{s}' = \psi \bar{s}'_p + (1 - \psi) \bar{s}'_r \quad (71)$$

9.3 Geographic Capital Distributions

This would quantify the amount of academic capital per capita, and per GDP produced at different geographic locations. This can also be split by the percentage contribution of academic fields towards the total academic capital of a region. This would inform us about the emergence of specific locations on the globe that would

be more specialized, as opposed to more generalists, and if there is a preference for one or the other based on regional resources or economic growth trajectories.

Consider $\Pi_x, \mathbf{x}_x$ be the portfolio, and the academic capital of an institute i in the geographic region θ . Then, the total capital of the region is simply a sum over $\mathbf{x}$ from each institution in the region.

$$\mathbf{x}_\theta = \sum_x \mathbf{x}_x \quad (72)$$

Without loss of generality, consider $\mathbf{x}_d$ be the academic capital from an academic field of depth $d \in D$, where d can be any depth, $d_{1,2,3,4} \in D_{1,2,3,4}$ then,

- $\mathbf{x}_\theta = \sum_D \mathbf{x}_d$, Total academic capital of a region should be equal to the sum of the academic capital split across all fields at a certain depth
- The proportionate contribution of d can be computed as $\mathbf{x}_d / \mathbf{x}_\theta$
- The concentration of academic capital contribution of a region can then be computed using Herfindahl-Hirschman index

$$\text{HHI}_D = \sum_{d \in D} \left(\frac{\mathbf{x}_d}{\mathbf{x}_\theta} \right)^2 \quad (73)$$

In addition, consider the per capita academic capital of a region θ to be

$$\mathbf{x}_{\text{PC},\theta} = \mathbf{x}_\theta / \text{pop}_\theta$$

Then, the inequality of academic capital contribution across regions can then be computed by the Gini coefficient, $Gini_{\text{PC}}(\Theta)$ Gini [1912], where all regions $\theta \in \Theta$ are ordered in increasing order of $\mathbf{x}_{\text{PC},\theta}$.

$$Gini_{\text{PC}}(\Theta) = \frac{1}{\mathbf{x}_\Theta} \sum_{x=1}^{|\Theta|} \sum_{y=1}^{|\Theta|} \text{abs}(\mathbf{x}_{\theta,y} \cdot \text{pop}_{\theta,y} - \mathbf{x}_{\theta,x} \cdot \text{pop}_{\theta,x}) \quad (74)$$

We can also compute the per capita over subset of contributors $\hat{C}_\theta$ from the region instead of the whole population,

$$\mathbf{x}_{\text{PC}',\theta} = \mathbf{x}_\theta / |\hat{C}_\theta|$$

, with Gini coefficient as,

$$Gini_{\text{PC}'}(\Theta) = \frac{1}{\mathbf{x}_\Theta} \sum_{x=1}^{|\Theta|} \sum_{y=1}^{|\Theta|} \text{abs}(\mathbf{x}_{\theta,y} \cdot |\hat{C}_{\theta,y}| - \mathbf{x}_{\theta,x} \cdot |\hat{C}_{\theta,x}|) \quad (75)$$

Similarly, the regions θ , can also be arranged in ascending order of per GDP academic capital, $Gini_{\text{GDP}}(\Theta) = \mathbf{x}_\theta / \text{GDP}_\theta$. Then the Gini coefficient is given by,

$$Gini_{\text{GDP}}(\Theta) = \frac{1}{\mathbf{x}_\Theta} \sum_{x=1}^{|\Theta|} \sum_{y=1}^{|\Theta|} \text{abs}(\mathbf{x}_{\theta,y} \cdot \text{GDP}_{\theta,y} - \mathbf{x}_{\theta,x} \cdot \text{GDP}_{\theta,x}) \quad (76)$$

While the Gini coefficients in (76) and (74) are defined across all academic fields, they can also be defined for each depth of the field as well. Combined with the HHI measures per region, and GINI coefficients globally, the metrics give a sense of concentration and inequality of academic capital per geographical region, sliced by academic fields.

A healthy system would be indicated by different regions of the globe contributing specialized knowledge to the global pool while retaining sufficient cross-pollination between the academic fields. An unhealthy system would be indicated by a concentration of academic capital in certain geographies across multiple academic fields, implying no specialization, while certain other regions do not contribute to the knowledge pool at all. Also, greater collaboration between specialized geographical locations would lead to greater contributions in academic capital made by that region, to domains out of its specialization.